

stats101a_hw4

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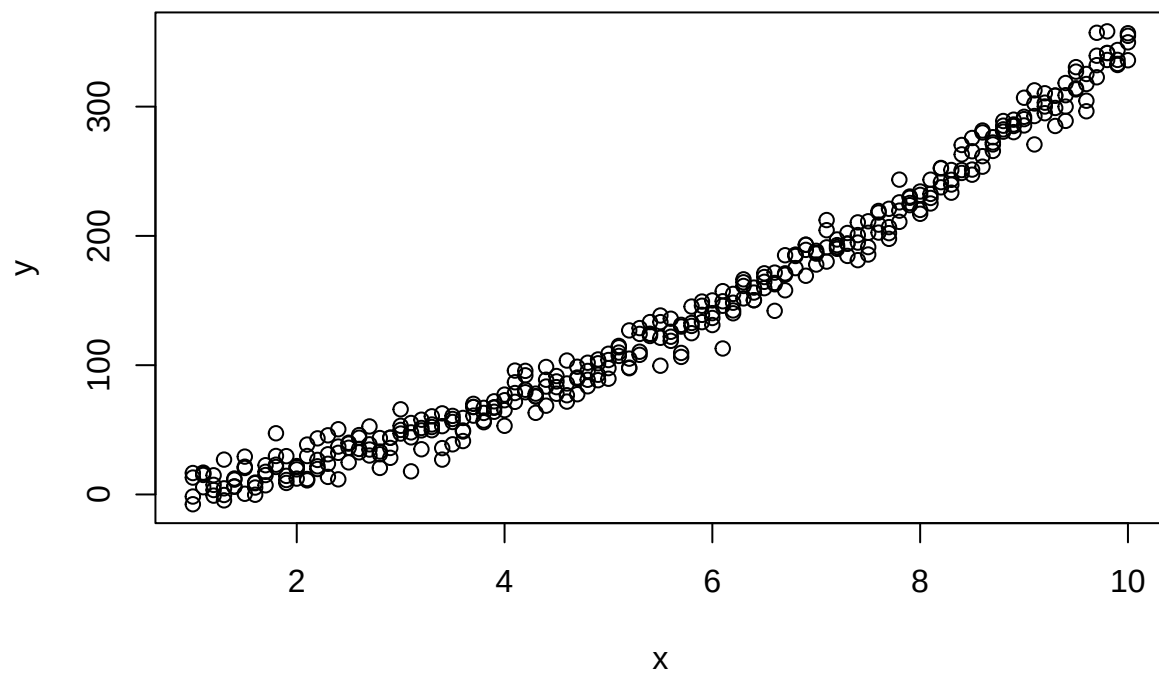
2026-04-23

```
generate_quadratic <- function(a, b, c, sigma, x = rep(seq(1, 10, 0.1), 4), random.seed) {  
  
  # Compute deterministic part  
  f_x <- a + b * x + c * x^2  
  
  # Generate noise  
  epsilon <- rnorm(length(x), mean = 0, sd = sigma)  
  
  # Generate response  
  y <- f_x + epsilon  
  
  y  
}
```

(1)

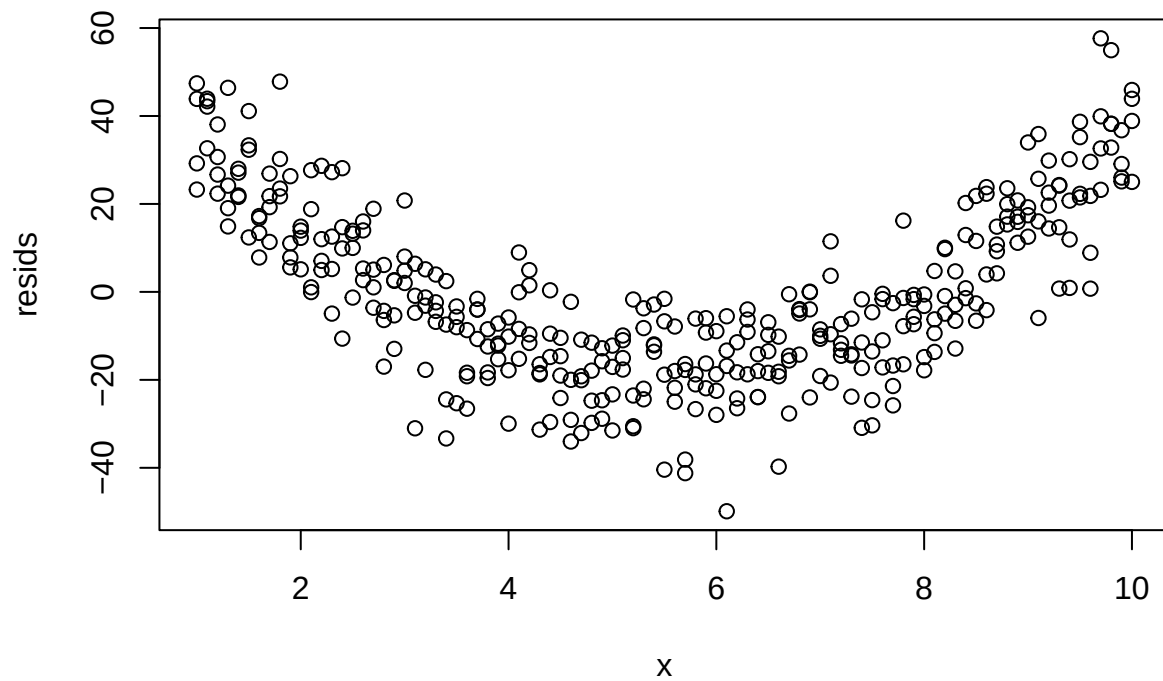
(a)

```
a = 1  
b = 5  
c = 3  
sigma = 10  
x = rep(seq(1, 10, 0.1), 4)  
y <- generate_quadratic(a,b,c, sigma, x, set.seed(123))  
  
plot(x, y)
```



(b)

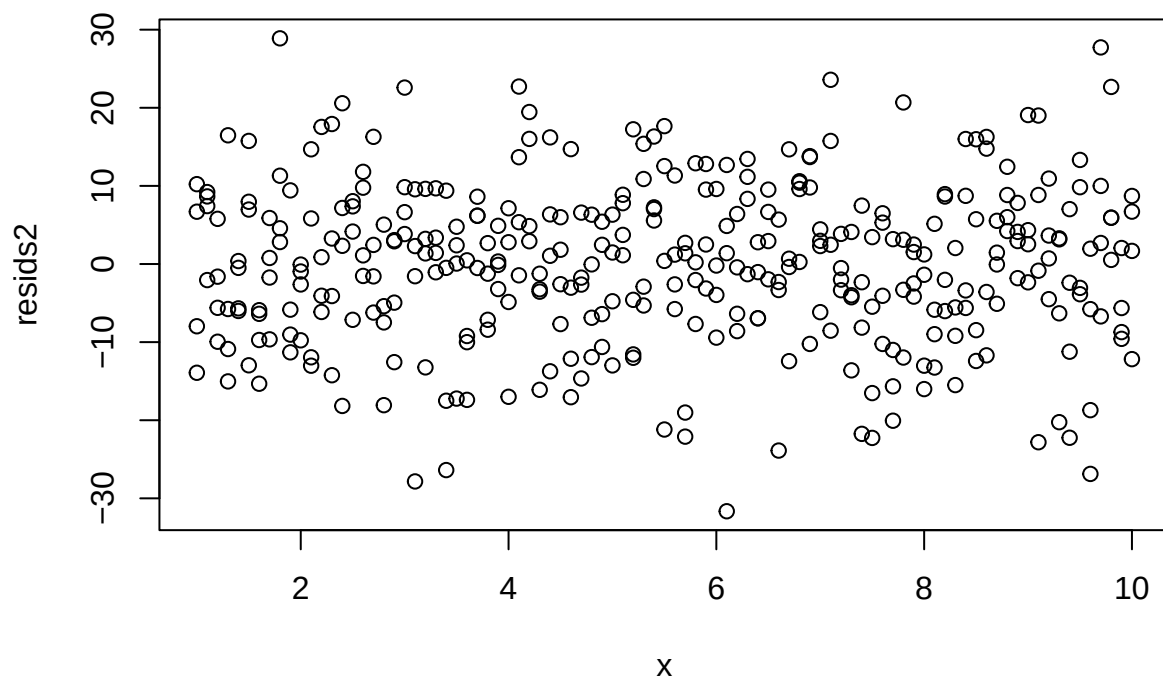
```
model_linear <- lm(y ~ x)
resids <- resid(model_linear)
plot(x, resids)
```



The points form a U-shaped (parabolic) pattern. positive at low x-values, negative in the middle, positive at high. systematic curved pattern. It is a non-random pattern. The key feature is the U-shaped (curved) pattern in the residuals.

(c)

```
model2 <- lm(y ~ x + I(x^2))  
resids2 <- resid(model2)  
plot(x, resids2)
```



Residuals are randomly scattered around zero with no visible pattern. no trends.

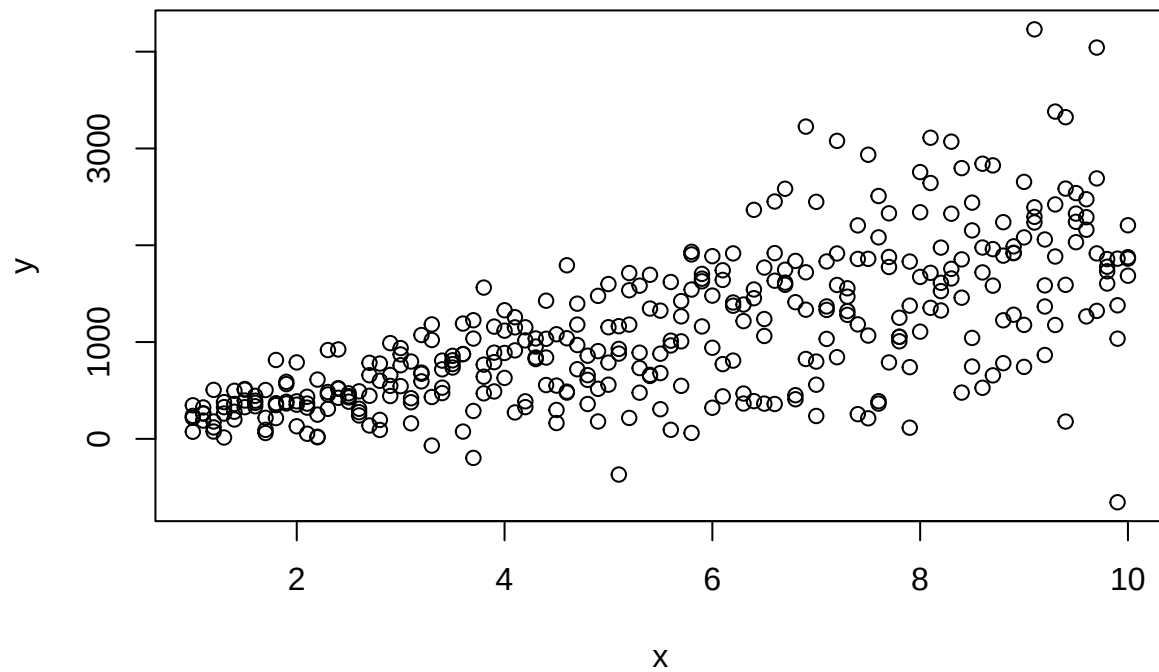
(d)

```
simulate_err <- function(a, b, x=rep(seq(1,10,.1),4), sigma, random.seed){
  a+b*x+rnorm(length(x),0,sigma*x)
}

x=rep(seq(1,10,.1),4)
a= 1
b=200
sigma=100
random.seed=123

y <- simulate_err(a,b,x,sigma,random.seed)

plot(x,y)
```

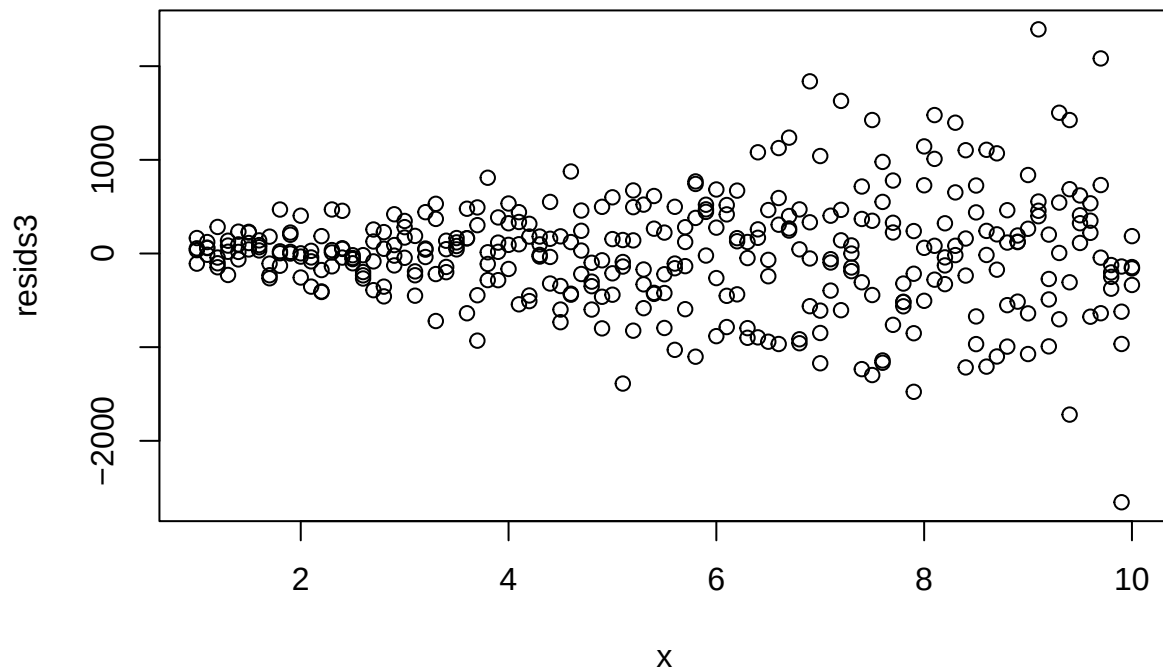


Errors have variance that increases with x . The variance is not constant (unlike the linear model with an error with the same variance). So the variance increases as x increases and the data becomes more and more spread out.

(e)

```
model3 <- lm(y ~ x)
resids3 <- resid(model3)

plot(x, resids3)
```



There's a cornucopia shape or fan-shape where the residuals increase in magnitude over x.

(2)

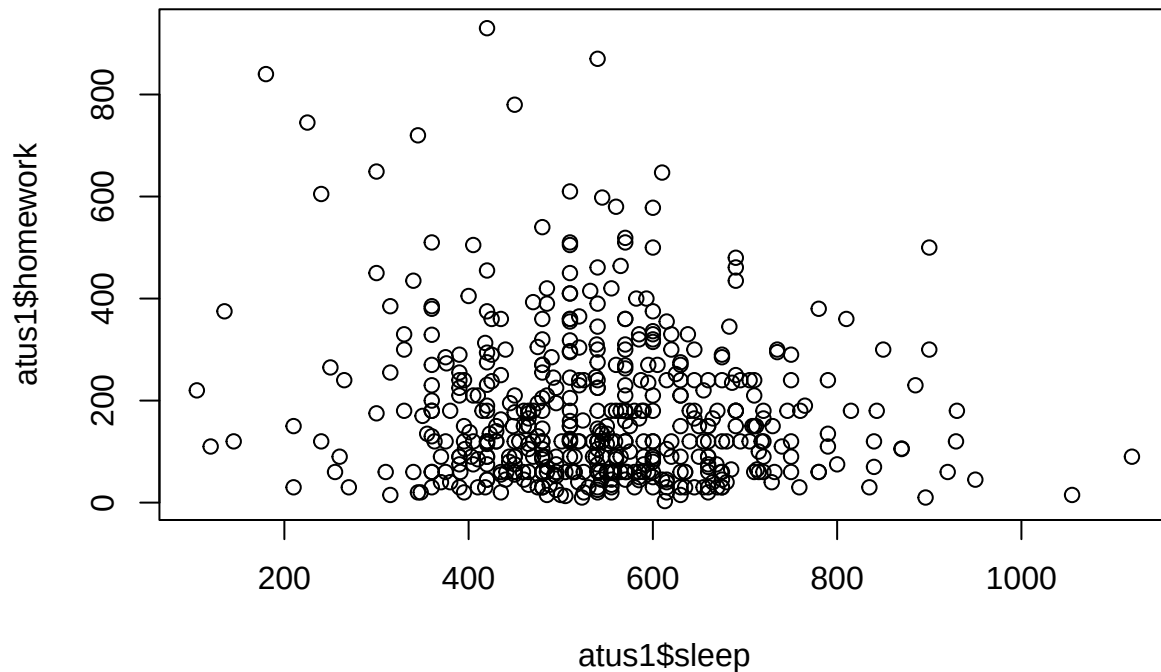
```
atus <- data.frame(read.csv('./atus.csv'))
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.2      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
atus1 <- filter(atus, homework > 0 )
```

(a)

```
plot(atus1$sleep, atus1$homework)
```



No the trend does not seem to be linear.

(b)

```
model <- lm(homework ~ sleep, data = atus1)
```

```
summary(model)
```

```
##
## Call:
## lm(formula = homework ~ sleep, data = atus1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -193.43 -108.59  -39.41   71.75  735.34
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 249.76178   26.00774   9.603  < 2e-16 ***
## sleep       -0.13120    0.04651  -2.821  0.00498 **
## ---
```

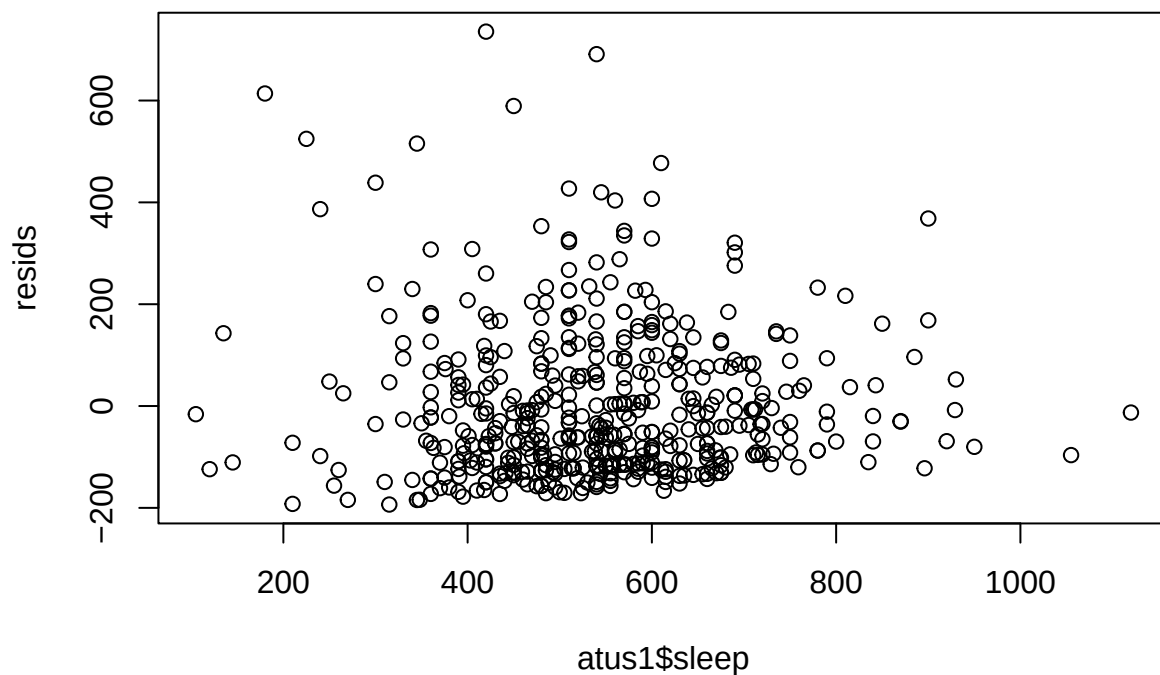
```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 146.2 on 493 degrees of freedom
## Multiple R-squared:  0.01589,    Adjusted R-squared:  0.01389
## F-statistic: 7.958 on 1 and 493 DF,  p-value: 0.004981
```

```
# i: no sleep = intercept = b0
# 249.76178

# ii: slope = b1
# -0.13120
```

(c)

```
resids <- resid(model)
plot(atus1$sleep, resids)
```



no clear pattern or trend since the residuals are scattered fairly randomly. no distinguishable pattern. This would suggest that the relationship between sleep and homework is roughly linear.

looks a bit like variance is not constant? the spread of residuals looks uneven.

outliers exist (some really large one)

(3)

Do Problem 1 in Chapter 2 of the textbook (on page 38).

```
df <- data.frame(read.csv('./playbill.csv'))

str(df)

## 'data.frame': 18 obs. of 3 variables:
## $ Production : chr "42nd Street" "Avenue Q" "Beauty and Beast" "Bombay Dreams" ...
## $ CurrentWeek: int 684966 502367 594474 529298 570254 319959 579126 134042 105853 822775 ...
## $ LastWeek : int 695437 498969 598576 528994 562964 282778 583177 152833 105698 836959 ...
```

(a)

```
model <- lm(CurrentWeek ~ LastWeek, data = df)
confint(model, level = 0.95)
```

```
##                2.5 %          97.5 %
## (Intercept) -1.424433e+04 27854.099443
## LastWeek      9.514971e-01   1.012666
```

The confidence interval is (9.514971e-01, 1.012666). 1 is a plausible value since it lies within the interval.

(b)

```
b0_hat <- coef(model)[1]
se_b0  <- summary(model)$coefficients[1, 2]

t_stat <- (b0_hat - 10000) / se_b0

p_value <- 2 * pt(-abs(t_stat), df.residual(model))
p_value
```

```
## (Intercept)
##      0.7517807
```

since the p-value is greater than 0.05, we fail to reject the null hypothesis of $\beta_0 = 10000$

(c)

```
new_data <- data.frame(LastWeek = 400000)
predict(model, newdata = new_data, interval = "prediction", level = 0.95)
```

```
##      fit      lwr      upr
## 1 399637.5 359832.8 439442.2
```

We are 95% confident that the actual gross box office this week will fall between 359832.8 and 439442.2 dollars for a single new production with \$400,000 last week.

\$450,000 is not a feasible value since it is not within the bounds set up by the 95% confidence interval.

(d)

This rule is very simple but ultimately too simplistic and sacrificing a lot of nuance and complexity. it may perform moderately well when week-to-week changes are small, but usually there will be a trend (downward or upward) that this rule won't reflect.